

ECON108 Winter 2025

Milestone 3

Seeun Park

Suggested adjustments:

- comparing linear vs polynomial functional form (age vs age and age2)
- comparing log vs linear functional form (lnincome vs income)
- comparing linear vs indicators (Num children vs any children)
- linear time trend vs year fixed effects
- changing the grouping of your indicator variables (indicators for republican, democrat instead of strong republican, moderate republican, moderate democrat, strong democrat)

Bonus points for the final project:

- interaction terms (female x education)
- If you have a binary dependent variable, estimating a probit model (note: you will have to report marginal effects in your regression table)

Tips from your TA

- Start with a simple linear probability model.
- Add more variables in Model 2 and continue refining the model based on the suggestions above.
- Don't forget to include five common variables: **race**- and **gender**-related dummies, as well as the continuous variables **educ**, **age**, and **year**.
- Be mindful of perfect collinearity.
- When interpreting a dummy, be mindful of your reference group. For example, if you're interpreting a **female** dummy, the estimate represents the difference in probability that $Y=1$ between females and males.
- We're doing economics, not just coding. Don't mindlessly run random equations. **Think through** what you're estimating, run regressions purposefully, and **teach yourself how to interpret** the results. This is where you apply the theories you've learned to real-world data—exactly what's expected of you as an economics major.

Research Question: I examine how racial stereotypes and political views influence individuals' perceptions of welfare spending.

Question 1. Model building

$$\textbf{Model 1: } \Pr(Welfare_i = 1) = \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + u_i \quad (1)$$

$Welfare_i$ is a binary variable that takes the value of 1 if individual i thinks the government's welfare spending is too much, and 0 otherwise.

$$\begin{aligned} \textbf{Model 2: } \Pr(Welfare_i = 1) = & \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i \\ & + \beta_4 \text{Conservative}_i + u_i \end{aligned} \quad (2)$$

In Model 2, I add political view dummies because political views may influence individuals' perceptions of specific races' work ethic as well as opinions on welfare spending. Omitting these variables could bias my estimates. Column (2) shows that controlling for political views slightly reduces the magnitude of the coefficients on the variables of interest, **White lazy** and **Black lazy**, indicating that the previous estimates in Model 1 capture the relationship between political views and the outcome variable. Additionally, because the political view dummies are statistically significant at the 5% level, I prefer Model 2 to Model 1.

Comment:

I included demographic characteristics separately in Model 3 because, in my case, adding them significantly changes the coefficient of one of my variables of interest, and I wanted to emphasize that. However, I recommend including all controls in Model 2 and making more interesting adjustments in Model 3.

$$\begin{aligned} \textbf{Model 3: } \Pr(Welfare_i = 1) = & \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i \\ & + \beta_4 \text{Conservative}_i + \beta_5 \text{Female}_i + \beta_6 \text{Black}_i + \beta_7 \text{OtherRace}_i + \beta_8 \text{Age}_i \\ & + \beta_9 \text{EduYears}_i + \beta_{10} \text{Year}_t + u_i \end{aligned} \quad (3)$$

In the second adjustment, I control for demographic characteristics that may influence both the variables of interest and the outcome variable. Adding these variables makes the coefficient on **White Lazy** statistically insignificant, suggesting that the previous estimate was driven by omitted demographic characteristics. Therefore, I prefer Model 3 over Model 2.

$$\begin{aligned} \textbf{Model 4: } \Pr(Welfare_i = 1) = & \beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i \\ & + \beta_4 \text{Conservative}_i + \beta_5 \text{Female}_i + \beta_6 \text{Black}_i + \beta_7 \text{OtherRace}_i + \beta_8 \text{Age}_i \\ & + \beta_9 \text{EduYears}_i + \sum_t \gamma_t \text{Year}_t + u_i \end{aligned} \quad (4)$$

,where $t \in \{1994, 1996, \dots, 2016, 2018, 2021, 2022\}$

In Model 4, I control for year fixed effects instead of a linear time trend...with 1990 as the base-line year... (*Explain the motivation for your adjustment. Decide whether you prefer the modified regression or not (look at p-values)*)

Model 5: $\Pr(Welfare_i = 1) = \Phi\left(\beta_0 + \beta_1 \text{WhiteLazy}_i + \beta_2 \text{BlackLazy}_i + \beta_3 \text{Liberal}_i\right.$
 $\left. + \beta_4 \text{Conservative}_i + \beta_5 \text{Female}_i + \beta_6 \text{Black}_i + \beta_7 \text{OtherRace}_i + \beta_8 \text{Age}_i\right.$
 $\left. + \beta_9 \text{EduYears}_i + \sum_{t=1}^T \gamma_t \text{Year}_t + u_i\right)$ (5)
 ,where $t \in \{1994, 1996, \dots, 2016, 2018, 2021, 2022\}$

When I check the predicted outcomes from Model 4, I find that a few observations fall outside the [0,1] range. Therefore, in Model 5, I estimate the same model using a probit specification. I prefer Model 4 for three reasons. First, ...

The Final Preferred Model ([Model #](#))

(*Write out the equation for your final preferred model here.*)

Question 2. Formatted table of regression output**Outcome:** Welfare Spending is Too Much (=1)

	(1)	(2)	(3)	(4)	(5)
	LPM	LPM	LPM	LPM	Probit
Variable name 1					
Variable name 2					
Variable name 3					
Variable name 4					
Variable name 5					
Variable name 6					
Variable name 7					
Age					
Years of Education					
Year					
Constant					
Observations					
R-squared					
Standard errors in parentheses					
*** p<0.01, ** p<0.05, * p<0.1					

(Your regression results)

This is an example table format. The Probit model does not have to be included in column (5), and you are not required to use a Probit model at all. Feel free to include more variables than those listed.

Question 3. Discussion of final model

For each qualitative variable:

- Interpret the estimated indicator coefficients correctly
- Discuss whether indicator coefficients are statistically significant (choose a significance level of 5% or 10%)
- Provide an economic explanation for your findings. What did you learn with respect to your research topic? What was surprising?

Qualitative variables

- **White lazy:**
- **Black lazy:** The probability that individuals think the government is spending too much on welfare is 0.105 (or 10.5 percentage points) higher for those who perceive Black people as lazy, compared to those who do not. This estimate is statistically significant at the 5% level.
- **Liberal:**
- **Conservative:**
- **Female:**
- **Black:**
- **Other Race:** Individuals of other races are 5.3 percentage points less likely to think that the government spends too much on welfare, compared to Whites, holding all else constant. This estimate is statistically significant at the 5% level.

To summarize, the results indicate that racial stereotypes, an individual's race, and political views are closely related to perceptions of government spending on welfare. This suggests that politicians who hold biased views about Black people's work ethics, are more conservative, and are White, are more likely to believe that government spending on welfare is excessive and seek to reduce the welfare budget, which could widen economic inequality. This finding is especially pertinent in the current political climate, where a conservative White president with racially biased views has taken office in the United States.

For each quantitative variable:

- If you have a polynomial, explain the shape of the relationship between the explanatory variable and the dependent variable. Find the turnaround point and discuss whether you think it's relevant.
- If you have a linear or log relationship, interpret the coefficient correctly
- Discuss whether coefficients are statistically significant (choose a significance level of 5% or 10%)
- Provide an economic explanation for your findings. What did you learn with respect to your research topic? What was surprising?

Quantitative variables

- **Age:**
- **Years of Education:** For each additional year of education, the probability that an individual believes welfare spending is too much increases by 0.2 percentage points, holding all other factors constant. The estimate is statistically significant at the 5% level.

Individuals with a higher level of education are more likely to think that the government's spending on welfare is excessive, while age is not related to this perception. Given that years of education are generally positively correlated with income, this finding is somewhat predictable.

(Please attach your do-file/R-script pdf)

Adjustment examples

1. A quadratic function of **Age**

- $Pr(Y = 1) = \beta_0 + \text{other variables} + \gamma_1 \text{Age} + \gamma_2 \text{Age}^2 + u$
- Marginal effect of **Age**: $\frac{\partial Y}{\partial \text{Age}} = \gamma_1 + 2\gamma_2 \text{Age}$
- This allows nonlinear relationship between **Age** and the outcome variable Y.
- The effect of age on Y to change at different values of age
- *Tip*: When interpreting, be sure to note whether the marginal effect increases or decreases with age.

2. Log of income

- Take the log of a “quantitative” income variable such as **coninc** and include **lninc** instead of **coninc**: `gen lninc = log(coninc)`
- Review the lecture notes to explain why you would use the log of income instead of raw income.
- In a linear probability model, apply the interpretation of a linear-log model.
- *Tip*: Since probit/logit models already assume a nonlinear relationship, I would use $\ln(\text{income})$ in a linear probability model.

3. Year fixed effects

- Use year fixed effects (FE) instead of linear time trend
- Linear time trend: `reg y x year`
- Year FE model: `reg y x i.year`
- Review the lecture notes to explain what year fixed effects do.
- *Stata Tip*: Your table can become messy with coefficients for each year. If you add the `drop(i.year)` option in the `outreg2` command, the table will not show the coefficients for the years.
- *Tip*: You can interpret the coefficient for just one year, but be careful to note that it represents the difference in $Pr(Y=1)$ between that year and the reference year.

4. Probit model

- Because the coefficient on x gives you how much z-score changes when x changes by one unit, not how much Y changes (aka. marginal effects), we need additional Stata command to report marginal effects

```
probit y x
margins, dydx(*) post
outreg2 using ...
```

- Review the lecture notes to explain why you would use a probit model.

5. Interaction terms

- Example Stata commands: (They are the same command lines.)
 - Option 1. `gen female_educ = female*educ`
`reg y x female educ female_educ`
 - Option 2. `reg y x i.female##c.educ`
`i.female##c.educ` is identical to `female educ i.female#c.educ`
- Why would you want to include the interaction term? Review the lecture notes.
- The correct interpretation is the key.

6. Changing the grouping of indicator variables

- For instance, I cleaned my `polviews` in two ways: (1) three categories (liberal; moderate; conservative) (2) five categories (extremely lib; lib; mod; cons; extremely cons). In this case, I could run
 - Model A: `reg y x lib cons`
 - Model B: `reg y x extlib lib cons extcons`

Extra: How do you write Year Fixed Effects in the equation?

Suppose your regression model use data from the following years 2000, 2002, 2004, 2006, 2008, and 2010.

1. A linear time trend

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 Year_t + u$$

, where *Year* is a continuous variable

2. Year Fixed Effects (Don't include both! Use either a linear time trend or the year FEs.)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \delta_2 2002_t + \delta_3 2004_t + \dots + \delta_6 2010_t + u$$

Of course you can rewrite it as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \sum_t \delta_t Year_t + u$$

, where $t = 2002, 2004, 2006, 2008, 2010$

$2002_t, \dots, 2010_t$ represent a set of year dummies. For instance, 2002_t equals one if the survey year is 2002 and zero otherwise.

There are six years in total, but Stata displays coefficients for only five year dummies. Why?

If we were to include all six years,

$$\delta_1 2000_t + \delta_2 2002_t + \delta_3 2004_t + \dots + \delta_6 2010_t$$

we would introduce **perfect collinearity**—just as we cannot include both female and male dummies in a regression. So we should drop one, and by default, Stata drops the first year as the reference category, in this case 2000. **The omitted year serves as the baseline.**

How do you check which years were used in the estimation? Can I just `tab year`?

- **Simply running `tab year` is not enough.** This command only lists all years present in the dataset, but some may not be used in the regression. For example, even if your dataset contains observations from 1994 to 2022, your estimation will only include the years where “all” variables have non-missing values. If one variable has data only for 2018, 2020, and 2022, the estimation will be based only on those three years. This is why we carefully selected variables in Milestone 1 to ensure at least five overlapping survey years.

Steps to check which years were used:

1. Run the regression with year fixed effects: `reg y x1 x2 i.year`
2. Generate predicted values: `predict yhat, xb`
3. Check the years included in the estimation: `tab year if yhat ~= .`

This command filters for non-missing predicted values, ensuring that only years actually used in the regression are displayed.

4. Compare the tabulated years with the Stata regression output. The regression will show coefficients for all but one year, which serves as the baseline. (Again, typically, the first year is omitted.)

I hope you don't think '*I don't want to use year fixed effects.*'